

13 EPC_Industry_Analytics_Assignment_Summer_2026

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1. Abstract

In this project I performed an exploratory data analysis on the EPC_Industry_Analytics_Demo.xlsx dataset which is basically about engineering, procurement and construction. The dataset had 1200 rows and 14 columns which were processed through various steps like data preprocessing, data cleaning, statistical summary, correlation analysis, outlier detection, group based aggregation, feature engineering, ranking analysis and conditional filtering. Along with this I also loaded the MNIST data and performed K-Means clustering on it and finally found the F1 Score which came out to be 0.7084155580116207.

2. Introduction

This project is about the EPC Dataset which means it is about engineering, procurement and construction. The dataset basically has various columns and rows which needed to be processed and analyzed. EPC projects are large scale projects which involve a lot of financial data, regional data and contractor information and analyzing this kind of data helps in understanding patterns like cost overruns, equipment performance and contractor efficiency which are really important for decision making.

Initially we performed basic exploration of the data by which we got to know that the data had 14 columns and 1200 rows. After understanding the structure of the data we moved ahead with preprocessing and cleaning it and then performed various kinds of analysis on it. Along with this the MNIST dataset was also used to understand and implement K-Means clustering and evaluate it using the F1 Score.

The entire project was implemented using Python and the lectures really helped in building the coding part and learning the ways in which machine learning works and implementing the algorithms on the data.

Topics I received training on during the first two weeks of internship:

- Python programming and data structures
- Data loading and basic exploration using Pandas
- Data preprocessing and data cleaning techniques
- Statistical analysis and descriptive statistics
- Data visualization using Matplotlib and Seaborn
- Correlation analysis and outlier detection
- Introduction to Machine Learning supervised and unsupervised learning
- K-Means Clustering and model evaluation using F1 Score

3. Project Objective

- To perform basic exploration of the EPC dataset and understand its structure in terms of rows, columns and data types.
- To perform data preprocessing and data cleaning by finding duplicate rows, null values and treating the missing values using median imputation.
- To perform statistical summary and correlation analysis on the dataset to understand the relationships between the numeric columns.
- To detect outliers in the data using the Interquartile Range (IQR) method.
- To perform group based aggregation, feature engineering, ranking analysis and conditional filtering on the dataset. And to load the MNIST data, apply K-Means clustering on it and evaluate the model using the F1 Score.

4. Methodology

4.1 Basic Exploration of Data

Initially we performed basic exploration of the data by which we got to know that the data had 14 columns and 1200 rows. This helped in understanding the overall structure of the dataset before moving ahead with any kind of processing.

4.2 Data Preprocessing and Data Cleaning

In this step we performed data preprocessing and data cleaning in which we found the number of duplicate rows and null values in the columns. After finding the missing values we treated the dataset by filling the null values or missing values with the median of the respective column. Median was used because it is not affected by extreme values and gives a better estimate for skewed data.

4.3 Statistical Summary

After cleaning the data we performed a statistical summary of the whole dataset which gave us a complete overview of all the numeric columns including measures like mean, median, standard deviation, minimum and maximum values.

4.4 Correlation Analysis

Then we performed correlation analysis on the data for finding the correlation between the numeric elements. This helped in understanding which columns were related to each other and to what extent.

4.5 Outlier Detection

We checked the outliers of the data by using the Interquartile Range (IQR) method for understanding the outlier values present in the dataset. Values which fell outside the range of $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ were considered as outliers.

4.6 Group Based Aggregation

Next we performed group based aggregation in which we grouped the data by its project IDs and found the mean, median and mode of the data so as to find the summarized data values at a project level.

4.7 Feature Engineering

Then we created a new feature by using the first numeric column and the second numeric column offset by 1 and by dividing these two values we performed feature engineering and added a new derived column to the dataset.

4.8 Ranking Analysis

After feature engineering we performed ranking analysis on the data to rank the records based on selected metrics and identify the top and bottom performers in the dataset.

4.9 Conditional Filtering

Then we performed conditional filtering which gave a value of (602, 16) meaning the filtered data had 602 rows and 16 columns. This filter was applied to keep only those rows whose values were strictly greater than the mean of the respective column.

4.10 MNIST K-Means Clustering

Finally we loaded the MNIST data and performed K-Means clustering on it. The model was then evaluated and we found the F1 Score evaluation for the K-Means clustering which came out to be 0.7084155580116207.

Tools and Technologies used: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook.

5. Data Analysis and Results

5.1 Basic Exploration

- Total Rows: 1200
- Total Columns: 14
- Key columns: Date, Project_ID, Region, Project_Type, Contractor_Tier, Planned_Budget_K, Actual_Cost_K, Budget_Variance_Pct, Sensor_Vibration_mm_s, Sensor_Temp_C, Efficiency_Score, Material_Price_Index.

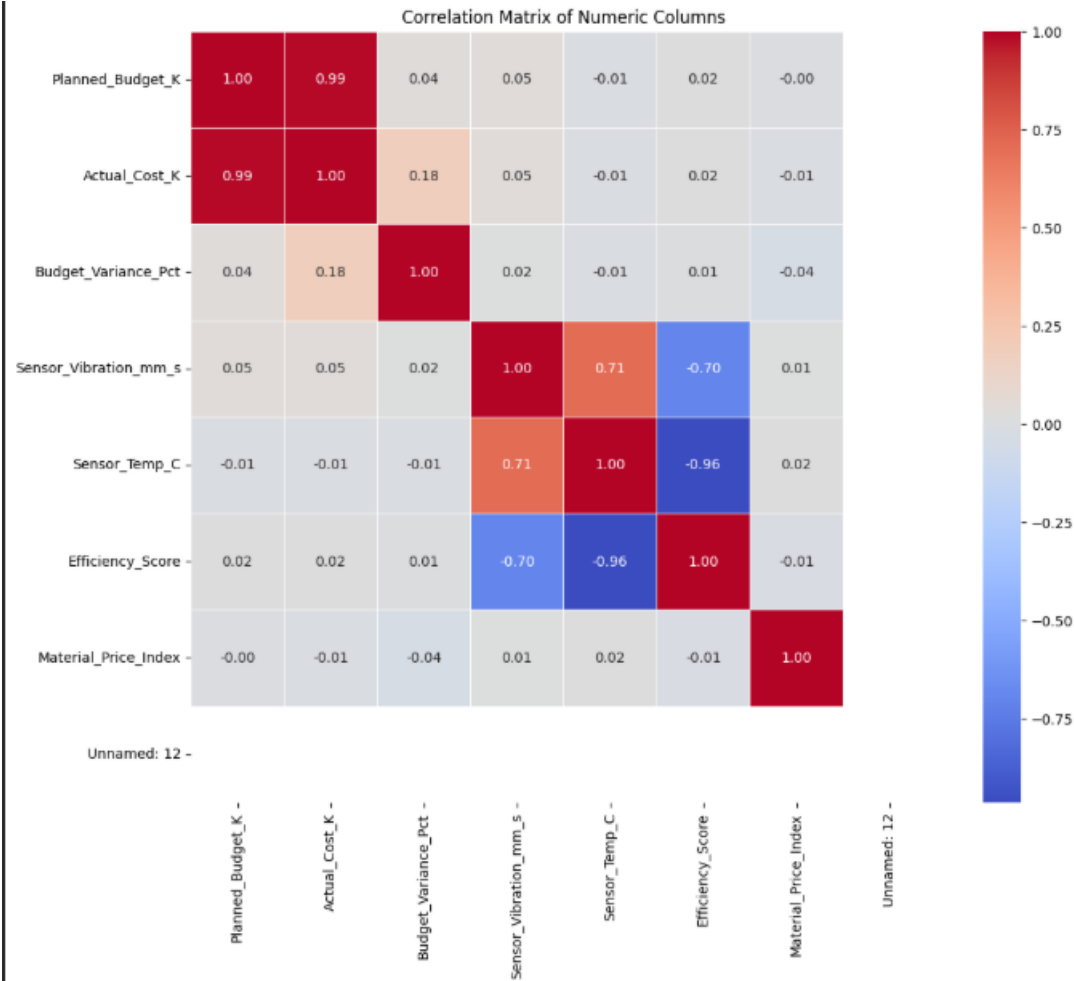
5.2 Data Cleaning Results

- Duplicate rows were identified and recorded.
- Null values were found in specific columns and were filled using the median of those columns.

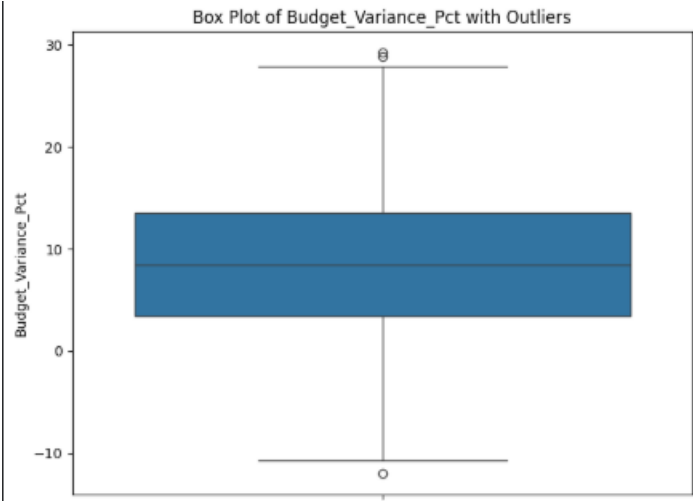
5.3 Statistical Summary

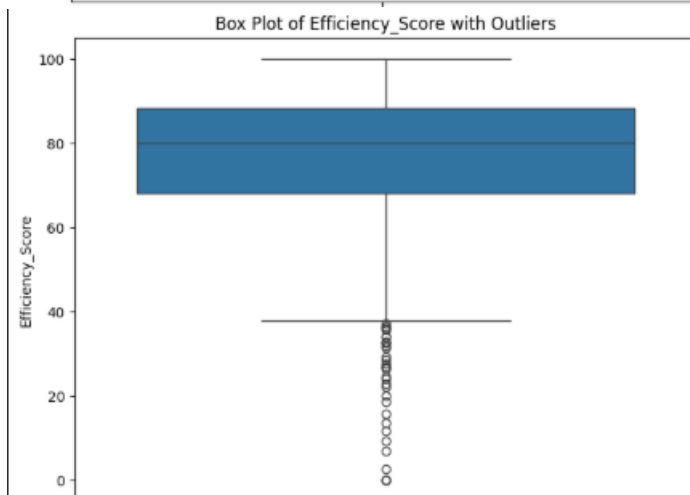
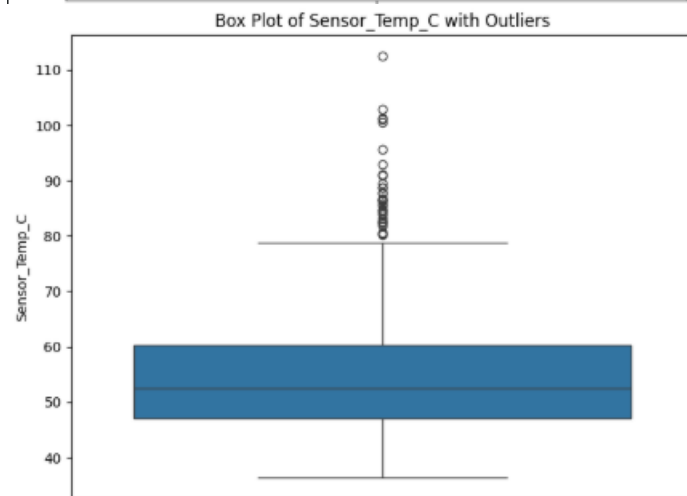
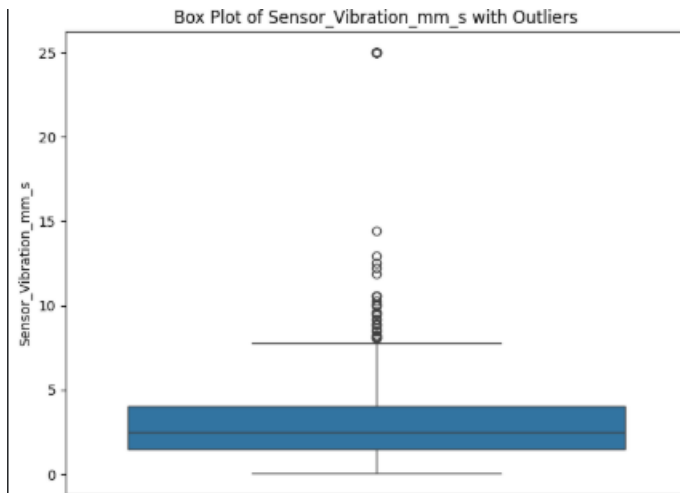
	Date	Project_ID	Region	Project_Type	\
count	1200	1200	1200	1200	
unique	NaN	1200	4	4	
top	NaN	PROJ-2183	LATAM	Renewable Energy	
freq	NaN	1	317	314	
mean	2023-05-30 12:00:00	NaN	NaN	NaN	
min	2023-01-01 00:00:00	NaN	NaN	NaN	
25%	2023-03-16 18:00:00	NaN	NaN	NaN	
50%	2023-05-30 12:00:00	NaN	NaN	NaN	
75%	2023-08-13 06:00:00	NaN	NaN	NaN	
max	2023-10-27 00:00:00	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	
	Contractor_Tier	Planned_Budget_K	Actual_Cost_K	Budget_Variance_Pct	\
count	1140	1200.000000	1200.000000	1200.000000	
unique	3	NaN	NaN	NaN	
top	Tier 2	NaN	NaN	NaN	
freq	574	NaN	NaN	NaN	
mean	NaN	2750.430575	2987.503387	8.482366	
min	NaN	500.052356	500.320284	-12.012275	
25%	NaN	1650.494262	1801.340050	3.374548	
50%	NaN	2751.711332	2967.673642	8.426896	
75%	NaN	3875.982343	4201.870148	13.529789	
max	NaN	4995.016754	6264.832318	29.201684	
std	NaN	1298.446505	1431.969622	7.294449	
	Sensor_Vibration_mm_s	Sensor_Temp_C	Efficiency_Score	\	
count	1200.000000	1200.000000	1200.000000		
unique	NaN	NaN	NaN		
top	NaN	NaN	NaN		
freq	NaN	NaN	NaN		
mean	3.131770	54.700500	76.433536		
min	0.045447	36.325602	0.000000		
25%	1.462936	47.091556	68.160112		
50%	2.473560	52.401493	80.128976		
75%	4.062142	60.197818	88.481005		
max	25.000000	112.475349	100.000000		
std	2.868782	10.514297	16.656041		
	Material_Price_Index	Unnamed: 12	Unnamed: 13		
count	1200.000000	0.0	1		
unique	NaN	NaN	1		
top	NaN	NaN			
freq	NaN	NaN	1		
mean	110.917631	NaN	NaN		
min	100.000000	NaN	NaN		
25%	105.460036	NaN	NaN		
50%	108.313578	NaN	NaN		
75%	118.146958	NaN	NaN		
max	121.113572	NaN	NaN		
std	6.464250	NaN	NaN		

5.4 Correlation Analysis



5.5 Outlier Detection





5.6 Group Based Aggregation

Planned_Budget_K			Actual_Cost_K		
Project_ID	mean	median	sum	mean	
PROJ-1000	2032.718750	2032.718750	2032.718750	2095.845956	
PROJ-1001	822.703500	822.703500	822.703500	839.771030	
PROJ-1002	2343.330027	2343.330027	2343.330027	2501.768017	
PROJ-1003	1900.478728	1900.478728	1900.478728	1966.592256	
PROJ-1004	3547.042879	3547.042879	3547.042879	3508.019596	

Budget_Variance_Pct					
Project_ID	mean	median	sum	mean	
PROJ-1000	3.105555	3.105555	3.105555	3.105555	
PROJ-1001	2.074566	2.074566	2.074566	2.074566	
PROJ-1002	6.761232	6.761232	6.761232	6.761232	
PROJ-1003	3.478783	3.478783	3.478783	3.478783	
PROJ-1004	-1.100164	-1.100164	-1.100164	-1.100164	

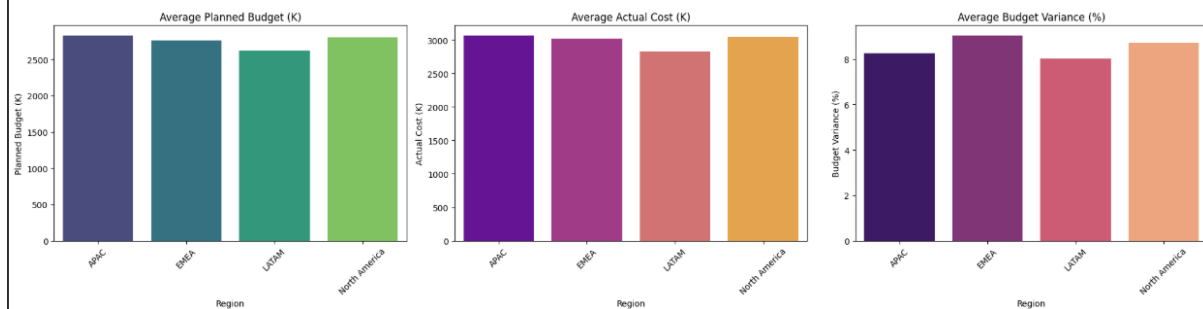
Sensor_Vibration_mm_s			Sensor_Temp_C		
Project_ID	mean	...	sum	mean	
PROJ-1000	4.775256	...	63.900729	64.144031	
PROJ-1001	1.779238	...	46.502858	83.031245	
PROJ-1002	2.303336	...	50.985558	81.513558	
PROJ-1003	3.341599	...	59.448200	72.523566	
PROJ-1004	3.501433	...	59.905765	70.703579	

Material_Price_Index					
Project_ID	mean	median	sum	mean	
PROJ-1000	100.000000	100.000000	100.000000	100.000000	
PROJ-1001	100.058382	100.058382	100.058382	100.058382	
PROJ-1002	100.116760	100.116760	100.116760	100.116760	
PROJ-1003	100.175133	100.175133	100.175133	100.175133	
PROJ-1004	100.233497	100.233497	100.233497	100.233497	

Unnamed: 12					
Project_ID	mean	median	sum	mean	
PROJ-1000	NaN	NaN	0.0	NaN	
PROJ-1001	NaN	NaN	0.0	NaN	
PROJ-1002	NaN	NaN	0.0	NaN	
PROJ-1003	NaN	NaN	0.0	NaN	
PROJ-1004	NaN	NaN	0.0	NaN	

[5 rows x 24 columns]

Average Financial Metrics by Region



5.7 Feature Engineering

A new derived feature was created by dividing the first numeric column by the second numeric column offset by 1 and this was added as a new column to the dataset.

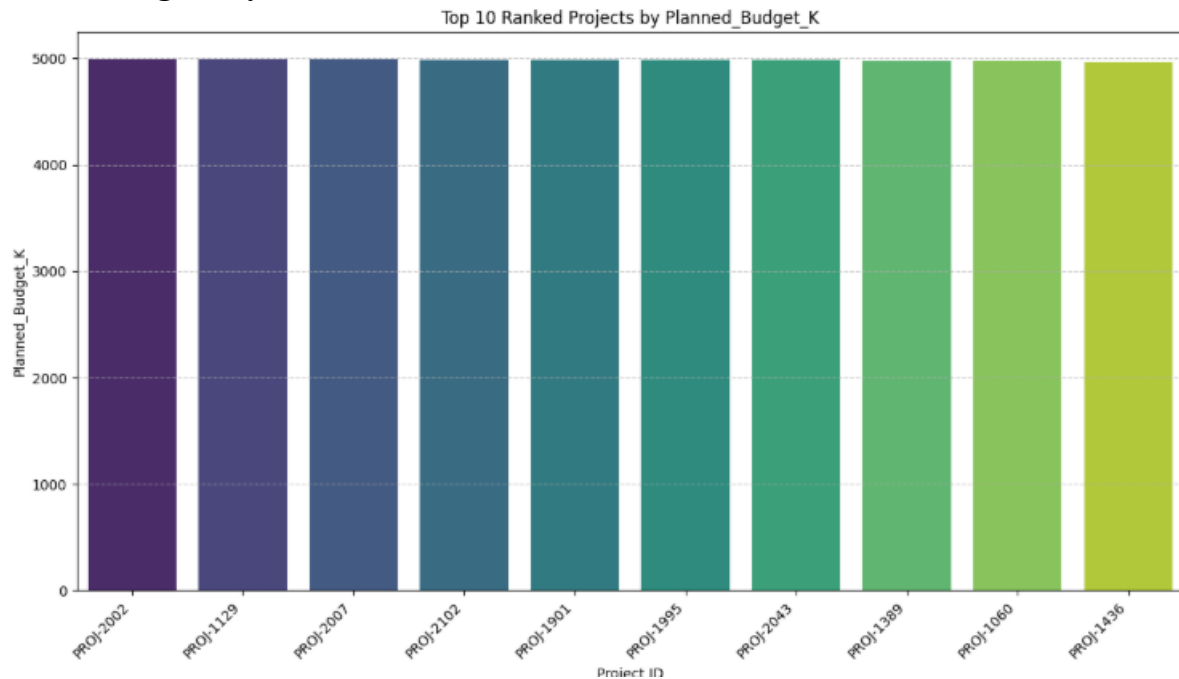
	Date	Project_ID	Region	Project_Type	Contractor_Tier	
0	2023-01-01	PROJ-1000	North America	Renewable Energy	Tier 3	
1	2023-01-01	PROJ-1001	LATAM	Industrial Plant	Tier 1	
2	2023-01-01	PROJ-1002	APAC	Oil & Gas	Tier 1	
3	2023-01-01	PROJ-1003	North America	Industrial Plant	Tier 2	
4	2023-01-02	PROJ-1004	North America	Renewable Energy	Tier 2	

	Planned_Budget_K	Actual_Cost_K	Budget_Variance_Pct	
0	2032.718750	2095.845956	3.105555	
1	822.703500	839.771030	2.074566	
2	2343.330027	2501.768017	6.761232	
3	1900.478728	1966.592256	3.478783	
4	3547.042879	3508.019596	-1.100164	

	Sensor_Vibration_mm_s	Sensor_Temp_C	Efficiency_Score	
0	4.775256	63.900729	64.144031	
1	1.779238	46.502858	83.031245	
2	2.303336	50.985558	81.513558	
3	3.341599	59.448200	72.523566	
4	3.501433	59.905765	70.703579	

	Material_Price_Index	Unnamed: 12	Unnamed: 13	New_Feature
0	100.000000	NaN	NaN	0.969417
1	100.058382	NaN	NaN	0.978511
2	100.116760	NaN	NaN	0.936295
3	100.175133	NaN	NaN	0.965891
4	100.233497	NaN	NaN	1.010836

5.8 Ranking Analysis

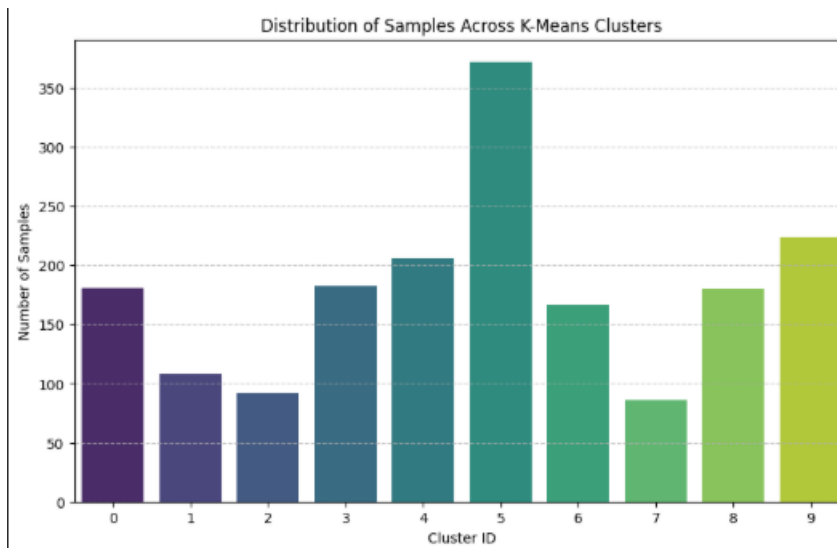


The records were ranked based on selected metrics which helped in identifying the top and bottom performers within the dataset.

5.9 Conditional Filtering

After applying conditional filtering where rows with values strictly greater than the column mean were retained, the resulting dataset shape came out to be **(602, 16)**.

5.10 MNIST K-Means Clustering Results



Metric	Value
Algorithm	K-Means Clustering
Dataset	MNIST Handwritten Digits
F1 Score	0.7084155580116207

6. Conclusion

So basically through this project I got to understand how to work with a real world dataset like the EPC Industry Analytics dataset end to end. Starting from basic exploration where we found that the data had 1200 rows and 14 columns, to cleaning it by handling null values and duplicates using median imputation, to performing statistical summary, correlation analysis, outlier detection using IQR, group based aggregation, feature engineering, ranking analysis and conditional filtering each step helped in understanding the data better.

The conditional filtering gave a resulting shape of (602, 16) which showed the records that were strictly above average. The K-Means clustering on the MNIST data gave an F1 Score of 0.7084155580116207 which shows a reasonably good performance for an unsupervised algorithm.

The lectures really helped in building the coding part of using Python and implementing the code as well as learning the ways in which machine learning works and implementing the algorithms on the data. In the future, supervised models like Random Forest or SVM could be applied to predict budget overruns in the EPC dataset and deep learning models could be used on MNIST for better accuracy.

7. APPENDICES

Appendix A — References

- Pandas Documentation: <https://pandas.pydata.org/docs/>
- Scikit-learn Documentation: <https://scikit-learn.org/stable/>
- MNIST Dataset: <http://yann.lecun.com/exdb/mnist/>
- Seaborn Documentation: <https://seaborn.pydata.org/>

Appendix B — GitHub Repository

<https://github.com/Vib167/Internship>